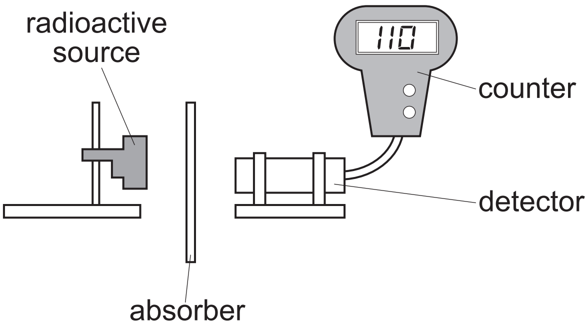
Examiner only

Answer **all** questions.

1. A teacher uses the apparatus below to demonstrate the penetrating properties of alpha, beta and gamma radiation.



(a) The teacher explains that there is a possibility of exposure to radiation from the source. Complete the risk assessment below. [2]

Hazard	Risk	Control measure
Nuclear radiation is ionising	....	....

(b) After the experiment the teacher gives the students some data about the radioactive source, cobalt-60, to analyse.
The data are given in the table below.

Absorber	Count rate (counts per second)
no absorber	256
paper	256
aluminium	110
lead	50

Use the data to answer the following questions.

(i) Explain how the data show that cobalt-60 does not emit alpha particles. [1]

....

....

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(ii) Explain how the data show that cobalt-60 emits beta and gamma radiation. [2]

....

....

....

(iii) The teacher tells the class that counts due to background radiation are included in the results in the table.

I. State **one** cause of background radiation. [1]

....

II. State how the results in the table should be corrected for background radiation. [1]

....

....

....

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